

MCA 5 Years
Semester – VIII
Data Mining & Warehousing(IC-811B)

Course Type: - Core Compulsory

Course Credits: - 4 Theory 0 Practical

Course Objective: - To understand data warehouses and data Mining with recent trends and development and trends in the field.

Course Outcomes – After completion of this course, learners will be able to

CO1 – Understand basic concepts of data warehousing and data mining.

CO2 – Learn preprocessing of data.

CO3 – Understand On Line Analytical Processing (OLAP)

CO4 – Learn data mining techniques and understand various algorithms and their applications.

CO5– Use of data mining as analytical tool in business data analysis.

Course Content:

Unit No.	Name	Contents	Hours
I	Introduction to Data Warehousing	Data Warehousing, Evolution of data warehousing systems, Data Warehouse (DW) - definition, Applications of DW, OLTP V/s Data Warehouse. Data Warehouse Architecture. Data Warehouse Server, Data Marts, Data Warehouse Implementation, Metadata. Advantages of Data Warehouse, Data Warehouse Pitfalls, Future of Data Warehousing. Data Warehouse Backend Process: Data Extraction, Data Cleaning, Data Transformation, Data Reduction, Data loading and refreshing.	8 Hrs
II	Online Analytical Processing (OLAP)	Multidimensional Data Model: Structuring/Modelling Issues, Derived Data, Schema Design, Dimension Tables, Fact Table, Star Schema, Snowflake schema, Fact Constellation, De-normalization. Introduction to OLAP, Strengths of OLAP, OLAP queries: Slicing and Dicing, Roll-up and Drill Down, OLAP Servers, Successful Warehouse, Presentation on ETL tool. Case Studies on Data warehousing	8Hrs
III	Data Mining	Fundamentals of data mining, Data Mining definitions, DBMS V/s Data Mining . Knowledge Discovery in Databases (KDD), Data Mining Functionalities, Issues and challenges in Data Mining. Data Mining Primitives, Data Mining Query Languages, Applications of Data mining, Data mining Case studies.	4 Hrs
IV	Association rule Mining:	Basic concept and terminology. Methods to discover association rules. A Priori algorithm. Partitioning, Pincer Search Algorithm, Dynamic Itemset Counting Algorithm, Applications of association rule mining.	8 Hrs

V	Clustering	Basic concept, measures of similarity, types of clustering methods. K-means algorithm. K-medoid algorithm, DBSCAN algorithm, applications of clustering.	4 H rs
VI	Classification:	Decision tress, basic concept, algorithm to build decision tress. Splitting methods in decision tress- GINI index and Information gain.	4 Hrs
VII	Web Mining and Spatial Mining	Web Mining; Web content mining, Web Structure mining, Web Usage mining, Text Mining, Temporal Mining and Spatial Data Mining.	4 Hrs

Books and Reading

1. Arun K Pujari, Data Mining Techniques, University Press
2. Jiawei Han & Micheline Kamber. Data Mining – Concepts and Techniques, Harcourt India
3. W. H. Inmon, Building the Data Warehouse. Wiley Dreamtech India Pvt. Ltd
4. Ralph Mikball. The Data Warehouse Life cycle Tool kit. Wiley Student Edition.
5. G. K. Gupta. Data Mining With Case Studies. Pearson education, India

Electronic Materials, Web Sites etc

1. http://akademik.maltepe.edu.tr/~kadirerdem/772s_Data.Mining.Concepts.and.Techniques.2nd.Ed.pdf
2. <http://infolab.stanford.edu/~ullman/mining/mining.html>

CO-PO Mapping

	PROGRAM SPECIFIC OUTCOMES	CO 1	CO 2	CO 3	CO 4	CO 5
PSO 1	Algorithm development	✓				
	Computer Based Solution	✓	✓	✓		
	Application Development	✓	✓	✓		
	Disciplinary Knowledge	✓				
PSO 2	Programming Skill		✓	✓	✓	
	Problem Solving/ Critical thinking	✓	✓	✓	✓	✓
	Mathematical Analysis	✓		✓	✓	
PSO 3	Financial Aspects					
	Modern Tools			✓	✓	
	Complex Problems solving	✓	✓	✓	✓	
	Time Frame Analysis					
PSO 4	Computer Architecture					
	Cooperation/Teamwork					✓
	Resource Management					
PSO 5	Modern Problem solving Technique	✓	✓		✓	
	Business Skills					✓
PSO 6	Communication skills					✓
	Decision making skills			✓	✓	✓
	Managerial Skill	✓		✓	✓	✓
	Environmental constraints					

	Research Skills	✓	✓	✓	✓	✓
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